DJI Agras – Troubleshooting Agras Drones 2022

Contents

1. The gasket/rubber pad needs to be replaced if the T40&T20P propellers are replaced

- **Notes:**

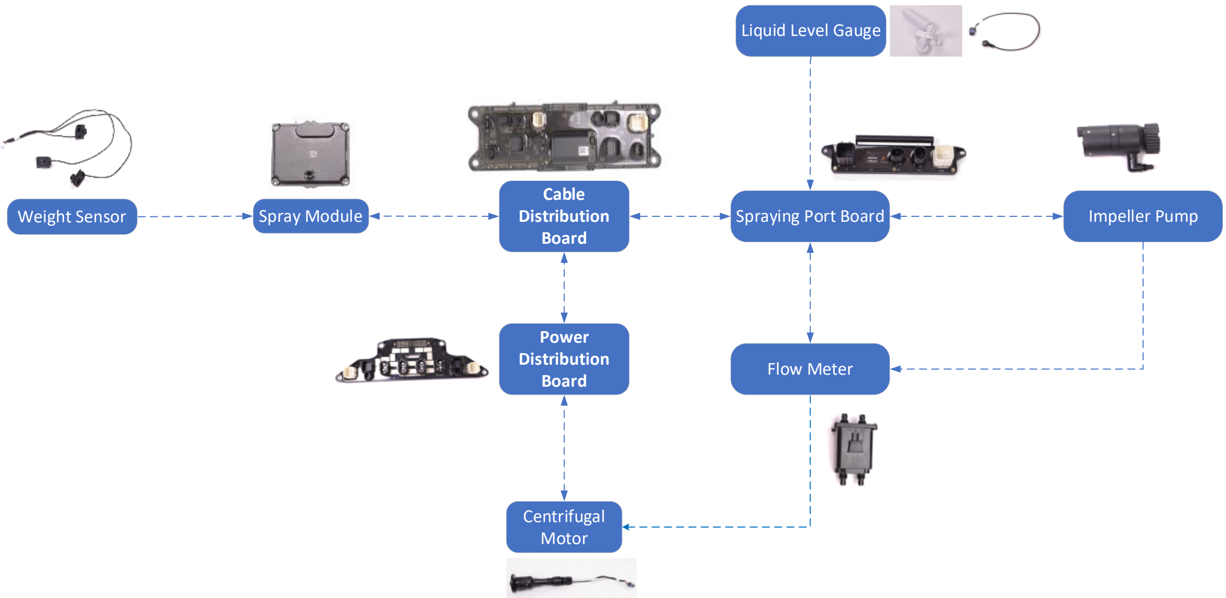
The gasket/rubber pad needs to be replaced if the T40&T20P propellers are replaced, as the **propeller gasket is a consumable part**. **After using the gasket for a long time or if the gasket is misshapen or damaged, the aircraft arm may vibrate abnormally, resulting in aircraft crash**. Therefore, it is necessary to replace the gasket together when replacing the propeller during the repair, and check whether the **gaskets on other propellers are misshapen**. If the gasket is not replaced after propeller replacement, which may result in the aircraft crash, and the dealer will be held accountable.

- **Related Materials:**

Teflon Gasket (YC.JG.MQ001292.03) (0.13mm; **two gaskets** on the upper part and two on the lower part)

Black Rubber Pad (YG.JG.MQ001280.03) (**one pad** on the upper part and one on the lower part)

2. Possible Causes and Solutions to T40&T20P Spraying System Remaining Liquid Payload Display Error



Issue 1: The remaining liquid payload is displayed as 0L when there is sufficient liquid payload in the spray tank

- **Cause of Error**

1. There is an error with the **liquid level gauge**.
2. There is an error with the **spraying board firmware** (occasionally).

- **Troubleshooting Steps**

1. Update the firmware of aircraft and remote controller to v01.00.0230.
2. If the remaining liquid payload value is large and a warning stating that there is an error with the liquid level gauge, it can be determined that the liquid level gauge is malfunctioning.
3. If the remaining liquid payload is displayed as 0L and no warning is shown, it can be determined that there is an error with the spraying board software.
4. If the remaining liquid payload is displayed normally after the aircraft is restarted or firmware is updated, it can be determined that there is an error with the spraying board firmware.

- **Solutions**

1. When the liquid level gauge is malfunctioning:
   i. **Clean the float of the liquid level gauge** (inside the spray tank);
   ii. Check whether the float of the **liquid level gauge and the bottom circuit board are installed in place**;
   iii. If the issue cannot be resolved after trying the methods above, **replace the sensor of the liquid level gauge**.
   iv. (It is best to update the aircraft firmware to v01.00.0230. In this case, the aircraft can continue to perform the mission even though the liquid level gauge is malfunctioning. But there will be large volume of liquid payload left in the spray tank (about 4L) after the operation is completed.
2. When there is an error with the spraying board firmware:
   i. Restart the aircraft multiple times.
   ii. If the issue cannot be resolved, try to update or downgrade the aircraft firmware.

Issue 2: When the spray tank is empty, the remaining liquid payload is displayed as 0.7L (T40) or 0.3L (T20P).

- **Cause of Error**

1. The spray **tank is pressed when it is empty**.
2. There is an **error with the flow meter or the flow meter is disconnected**.

- **Troubleshooting Steps**

1. If the flow meter value is shown 0 during operation, it can be determined that there is an error with the flow meter or the flow meter is disconnected.
2. If the aircraft firmware is v01.00.0230 and the flow meter is functioning normally, it can be determined that the spray tank is pressed when it is empty.

- **Solutions**

1. When there is an error with the flow meter: Check whether **the flow meter is connected properly**. If not, remove and plug it again. If the flow meter is connected, replace it.

2. If the spray tank is pressed when it is empty: **Restart the aircraft**.

Issue 3: When there is large volume of liquid in the spray tank, the remaining liquid payload is displayed as 0.7L (T40) or 0.3L (T20P).

- **Cause of Error**

The **liquid level gauge** is functioning normally at first and then it cannot work properly.

- **Solutions**

1. **Clean the float of the liquid level gauge** (inside the spray tank);
2. Check whether the float of the liquid level gauge and the **bottom circuit board are installed in place**;
3. If the issue cannot be resolved after trying the methods above, **replace the sensor of the liquid level gauge.**
4. It is best to update the aircraft firmware to v01.00.0230. In this case, the aircraft can continue to perform the mission even though the liquid level gauge is malfunctioning. But there is large volume of liquid payload left in the spray tank (about 4L) after the operation is completed.

Issue 4: When the spray tank is empty, the remaining liquid payload is displayed as 0.71L (T40) or 0.31L (T20P).

- **Cause of Error**

1. The float of the liquid level gauge cannot fall lower due to the adsorption of impurities;
2. The **liquid level gauge is short circuited**.

- Troubleshooting Steps

1. Clean the float of the liquid level gauge and fill the spray tank with approximately 3L of water. Then start spraying. If the remaining liquid payload is 0, it can be determined that the float of the liquid level gauge cannot fall lower due to the adsorption of impurities.
2. If the remaining liquid payload is 0.71L (T40) or 0.31L (T20P) after spraying is enabled, and the value does not change even though the spray tank is empty, it can be determined that the liquid level gauge is short circuited.

- **Solutions**

1. If the float of the liquid level gauge cannot fall lower due to the adsorption of impurities: **Clean the float of the liquid level gauge** (inside the spray tank);
2. If the liquid level gauge is short circuited: **Replace the sensor of the liquid level gauge**.

Issue 5: When there is large volume of liquid in the spray tank, the remaining liquid payload is displayed as 0.71L (T40) or 0.31L (T20P).

- **Cause of Error**

1. There is an error with **Tare Calibration**;
2. The **weight sensor cable is disconnected: One of the weight sensor's green signal cable is broken**.

- **Troubleshooting Steps**

1. Check whether the weight sensor cable is broken. If yes, it can be determined that the cable is disconnected; Otherwise, there is an error with Tare Calibration;

2. Open the Y-tee valve at the bottom of the spray tank and drain out the water in the tank. Then perform Tare Calibration multiple times.
3. After Tare Calibration is performed, fill in the spray tank with water, and the remaining liquid payload is displayed normally. Then it can be determined that there is an error with Tare Calibration.

- **Solutions**

1. If there is an error with Tare Calibration: Open the Y-tee valve at the bottom of the spray tank and drain out the water in the tank. Then **perform Tare Calibration multiple times**, fill in the spray tank with water of 20L (T40) or 6L (T20P), and perform **Tare Calibration** again.
2. If the weight sensor cable is disconnected: **Replace the weight sensor**.

Issue 6: The remaining liquid payload value keeps changing.

- **Cause of Error**

1. After the spraying board is replaced, Tare Calibration is performed when the weight sensor value fails to be written.
2. The screw securing the weight sensor base is loose.

- **Troubleshooting Steps**

1. If the spraying board is replaced, there is a high probability that the **weight sensor value fails to be written.**
2. The drone does not take off and the remaining liquid payload value only changes slightly, while the value keeps changing after takeoff, then it can be determined that the screw securing the weight sensor base is loose.

- **Solutions**

1. After the **spraying board is replaced**, Tare Calibration is performed when the weight sensor value fails to be written.
   i. **Write in the weight sensor value** within 70s after the drone is powered on. Note that you cannot click to reset the weight sensor value.
   ii. Open the Y-tee valve at the bottom of the spray tank and drain out the water in the tank. Then perform **Tare Calibration multiple times**.
2. If the screw securing the weight sensor base is loose: **Tighten the screw**.

Issue 7: The remaining liquid payload is shown as 59.5L.

- **Cause of Error**

1. After the spraying board is replaced, the weight sensor value fails to be written.
2. There is an error with the spraying board firmware (occasionally).
3. The weight sensor cable is disconnected: One of the weight sensor’s white signal cable is broken.

- **Troubleshooting Steps**

1. Check whether the weight sensor cable is broken. If yes, it can be determined that the cable is disconnected; Otherwise, there may be other factors.
2. If the issue occurs after the spraying board is replaced, the weight sensor value fails to be written.
3. If the issue occurs during operation, there is an error with the spraying board firmware.

- **Solutions**

1. After the spraying board is replaced, the **weight sensor value fails to be written.**
   i. Write in the weight sensor value within 70s after the drone is powered on. Note that you cannot click to reset the weight sensor value.
   ii. Open the Y-tee valve at the bottom of the spray tank and drain out the water in the tank. **Then perform Tare Calibration multiple times**.
2. When there is an error with the spraying board firmware:
   i. Restart the aircraft multiple times.
   ii. If the issue cannot be resolved, try to update or downgrade the aircraft firmware.
3. If the weight sensor cable is disconnected: Replace the weight sensor.

3. T40 & T20P FPV Camera Screen Black out After Unboxing

- **Issue Description:** The T40 & T20P **FPV camera's screen is black out** after unboxing.
- **Solution:**

Try to **update the firmware** to v01.00.0230 as FPV camera calibration data download may fail (small probability).

Check whether the **FPV camera cable** is attached properly.

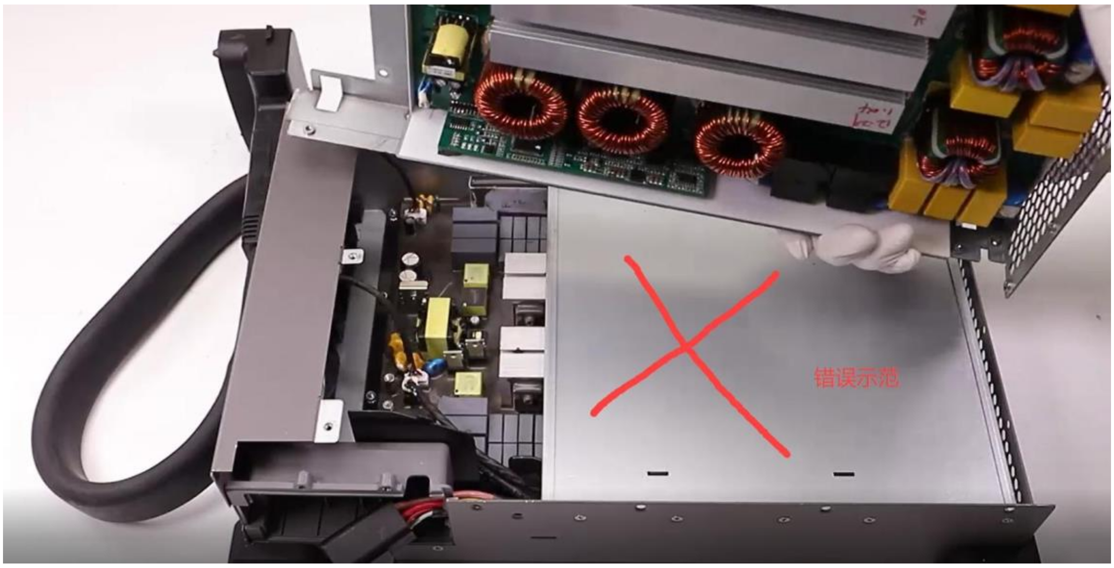
If the issue persists, check whether the **FPV camera and aerial-electronics board** are functioning or not.

4. T40&T20P Cable Distribution Module Replacement Notices

If the cable distribution module, aerial-electronics module, or RF module is replaced, the corresponding **thermal conductive pad** needs to be replaced and the location of the conductive pad must be cleaned.


5. The isolation plate needs to be removed when mounting AC module to DC Module

Please remove the isolation plate marked in the X as shown in the figure below when mounting AC module to DC module of the Intelligent Charger, avoiding the risk of electric shock or AC module scorched.



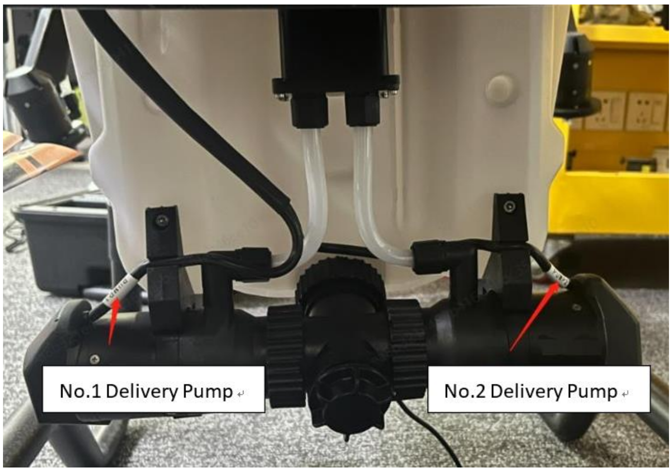
6. **What should I do if the T40/T20P remote controller prompts “Other errors”?**

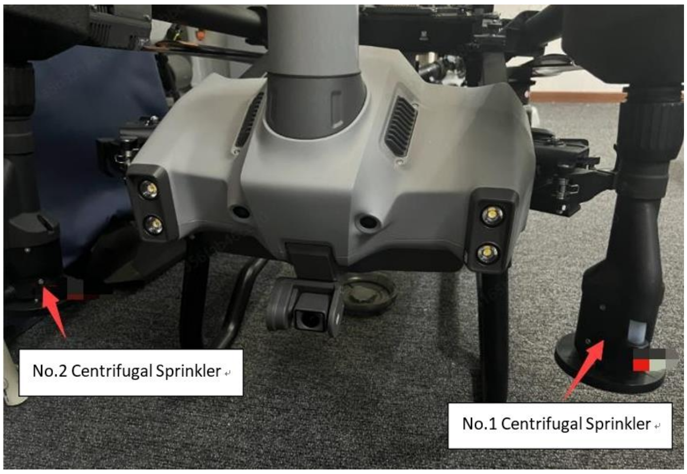
- **Issue Description: The app prompts “Other errors”.**
- **Solution:**

- Restart the aircraft. If the warning disappears, it can be used normally.
- If the issue occurs every time when the drone is powered on, it is recommended to **replace omnidirectional radar and backward radar.**

7. T40/T20P Delivery Pumps and Centrifugal Sprinklers Installation Position

- **As shown below:**





8. T40/T20P Power Distribution Board Maintenance Instructions

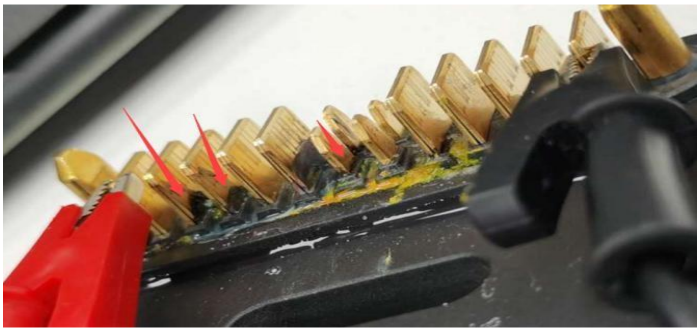
- **Maintenance Frequency and Content:**

1. Before operation, please check whether **chemical fertilizers**, **pesticide** and other **residues are**

accumulated at the connector of the power distribution board. If the residues have accumulated for a long time, **it is easy to cause a short circuit** of the power distribution board when the drone is turned on and there are risks in using the drone.

During repair, please clean the connector of the power distribution board before powering on the drone.

2. **Daily inspection** and maintenance are also required.



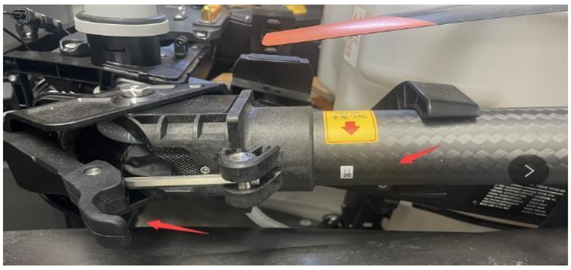
- How to clean the connector:

Wipe with a **cotton swab or soft toothbrush** with more than **95% alcohol to clean the connector**.

9. T40/T20P Aircraft Arm Inspection Required Items

- **Check the root of the aircraft arm.**
- **Operating Steps and Inspection Frequency:**

1. Check whether the **rubber sleeve** of the locking part is broken when unfolding the aircraft arm or during repair, and check whether the **contact surface between the locking part and the aircraft** is worn, cracked or underlayer material **is exposed**.



- **Solution:**

1. If the rubber sleeve of the locking part is broken, please replace it immediately. Otherwise, the **aircraft**

arm may be damaged by the locking part.

2. If the root of the aircraft arm is worn, cracked or underlayer material is exposed, **please replace the aircraft arm**.

- **Check the motor fixing base.**

Operating Steps and Inspection Frequency:

1. During repair, check whether the **motor fixing base is loose**, whether there is a gap in the slot of the ESC bracket, and whether the **fixing slot of the motor base is worn or cracked**.

- Solution: If the issues mentioned above occur, please **replace the corresponding aircraft arm**.

10. T40/T20P Repair Required Items

- Check whether the **propeller is bent**. The material of the T40/T20P propeller is thick, tough, and not easy to break. But the propellers may be bent after the drone crashes. Without replacing the propellers, the drone may **vibrate abnormally, causing the drone crash.**
- **Method**:
- Visually inspect the propellers:

Find a new propeller, place the propeller to be tested under the new propeller. The propeller is bent if there is a gap between two propellers. The propeller is qualified when there is a perfect fit for the two propellers.

- Recommended: **Take a ruler of 30 cm** or above (or other **straight** object), place the ruler to one side of the propeller, and fit the top of the ruler to the top of the propeller. Then check whether there is a gap between both sides of the ruler and propeller.

There is a large gap between the propeller and ruler as shown below, which can be determined that the propeller is bent. In this case, replace the propeller.

There is a small gap between the propeller and ruler as shown below, which can be determined that the propeller is not bent.

- If the propeller is bent, please replace it as soon as possible. After repair, perform flight testing and take off the drone with full payload, make the drone hover in place or fly to the left or right. Then check whether there are abnormal propeller sounds.

2. Check whether the propeller gasket/rubber pad is damaged.

- **Method**:

1. Check whether the order of propeller gaskets is correct: From top to bottom, **black pad x1 + white gasket x2 + propeller + white gasket x2 + black pad.**
2. If the propeller is replaced, make sure all gaskets and pads are replaced.
3. During repair, check whether the black rubber pad is misshapen. If yes, replace it immediately. Otherwise, the drone may vibrate abnormally, causing the drone crash.

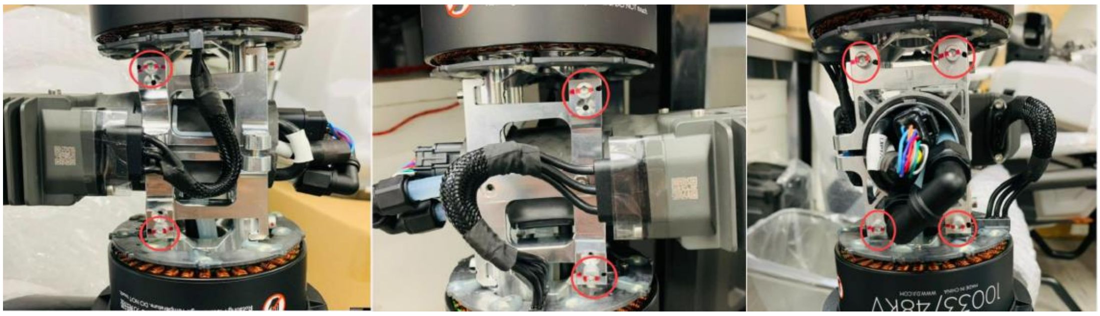
- **Check whether the motor base is loose.**

- **Method**: Check the four motor bases are loose.

First, hold the upper and lower motors with your hands, turn the motors counterclockwise and clockwise respectively, and check whether the motor rotates. If yes, use the dedicated tool to tighten the screws securing the motor base.

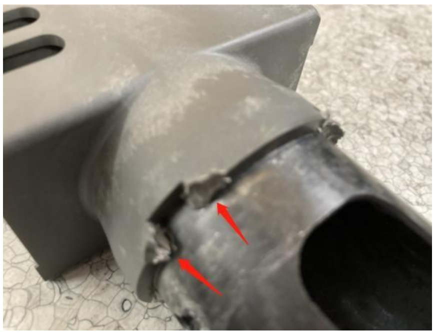
- **Check whether the screw securing the motor is broken.**
- **Method**:

1. Detach the four motor protection foam pads respectively, and check whether the screws are missing. If necessary, replace the screws. After the assembly is completed, **use a red pen to mark them**.
2. Use the electric screwdriver to tighten the screws and confirm that the screws are secured. If the screw is broken, replace the screws. After the assembly is completed, use a red pen to mark them.



- **Check whether the aircraft arm is broken.**
- **Method**:

1. First, check whether the **motor base limit slots** (as shown in the arrow) of the four aircraft arms are misshapen or broken. If yes, the corresponding aircraft arm needs to be replaced.



2. Check whether the **inside of the aircraft arm is broken**. If yes, the corresponding aircraft arm needs to be replaced.

- **Check whether the motor is broken.**
- **Method**: If the motor terminal surface (as shown in the figure) is cracked or damaged, replace it immediately, which can prevent liquid from corroding the motor and cause the spontaneous combustion.As shown below: The motor cracks (left figure) and the motor is broken (right figure). In this case, the motor needs to be replaced.